



MUHAMMAD AIMAN HAKIM BIN MOHD NIZAM

Address: Sungai Siput (U), Perak
Phone: 011-51468013
Email: hakimnizam772@gmail.com

PROFESSIONAL SUMMARY

Detail-oriented Software Engineer specializing in backend architecture, enterprise system modernization, and GIS integration. Proven track record in reverse-engineering legacy systems without source documentation and rebuilding scalable public-sector and enterprise applications. Expert in optimizing relational databases (PostgreSQL/MySQL) and building robust services using Laravel, native PHP, and Node.js ecosystems.

WORK EXPERIENCE

Programming, IT Madani Expert Sdn Bhd Jan 2026 - Present

- Involved in the development and maintenance of procurement and contractor management systems using Vanilla PHP for Lembaga Air Perak (LAP), including integration with legacy modules and existing workflows.
- Independently designed and architected a nationwide Tadika Alumni & Management System from the ground up using Laravel 12 and MySQL, implementing secure role-based access control via Laravel Breeze and dynamic reporting tools using Maatwebsite/Excel.
- Managed the full software development lifecycle, including database design, system analysis, debugging, testing, and deployment.
- Conducted intensive system analysis and reverse-engineering on a legacy public-sector PLGS system without original source code; successfully recovered critical business logic by map-analyzing live database structures and initiated the redevelopment using Laravel 12, Postgres 18, and advanced geospatial libraries (Geoserver, OpenLayers, Proj4js).

Junior Software Engineer, Carsome (K-Youth Program) May 2025 - Oct 2025

- Designed, developed, and maintained scalable backend services using JavaScript, Node.js, Express, and NestJS.
- Leveraged Git, GitHub, and GitLab CI/CD pipelines to streamline collaborative sprint development, implement automated testing environments, and ensure smooth production deployments.
- Utilized Jira actively for sprint planning, backlog refinement, and daily task tracking, increasing overall delivery transparency within a fast-paced Agile/Scrum team environment.

Internship, Novutal Consulting Sdn Bhd Sep 2024 - Jan 2025

- Developed and enhanced CRM and HRM modules in Odoo (lead management, sales tracking, biometric attendance).
- Designed workflow diagrams and system documentation to improve usability and training.
- Assisted in website development using HTML, CSS, and JavaScript, ensuring responsive and user friendly design.

EDUCATION

Bachelor of Computer Science with Honours (Networking) 2021 - 2025
Universiti Teknologi MARA, Shah Alam

Matriculation in Computer Science 2019 - 2021
Kolej Matrikulasi Perak

PROJECTS

Payung – Insurance Agent Directory & CRM System (Laravel | MySQL)

- Architected a dual-facing directory and CRM portal using Laravel 13.7 (PHP 8.3), Laravel Fortify/Jetstream, and Spatie for robust multi-tenant role security and multi-factor authentication.
- Reactive Frontend: Engineered a highly responsive frontend utilizing Livewire 4.3, Flux UI, and Tailwind CSS 4.0 to handle complex, real-time customer lead routing and quotation interfaces without SPA overhead.
- Automated Data Pipelines: Developed automated background schedulers to track policy expiration matrices for preemptive agent alerts, and built secure bulk data-ingestion pipelines for high-volume spreadsheet migrations.

PLGS System Modernization & Re-engineering (Laravel | Postgres | Reverse Engineering)

- Reverse-engineered a critical legacy PLGS system without access to original source code by map-analyzing complex live database structures.
- Architected and redeveloped the modernized base system using Laravel 12 (PHP 8.2), Postgres 18, and a Bootstrap 5 frontend powered by the Dayone Spruko template.
- Integrated advanced mapping and GIS features using Geoserver, OpenLayers v10.7.0, and Proj4js for handling precise coordinate systems (e.g., GDM2000).
- Implemented secure authentication using Laravel Fortify alongside fine-grained Access Control and audit trails via Spatie Role, Permission, and Activity Log.

SPEED System & SPEED Contractor System (LAP) (Vanilla PHP | MySQL)

- Contributed to the development and rigorous maintenance of core procurement and contractor management systems for Lembaga Air Perak (LAP).
- Implemented custom backend modules, complex database integrations, and legacy workflow enhancements using Vanilla PHP.
- Assisted in deep troubleshooting, debugging, and maintaining high-availability production systems to ensure operational stability.

Tadika Alumni & Management System (Laravel | MySQL)

- Built a comprehensive alumni management platform using Laravel 12 (PHP 8.2), featuring role-based authentication ('admin' and 'alumni') via Laravel Breeze.
- Implemented a dynamic frontend utilizing Blade templates, Tailwind CSS, Bootstrap 5, and Alpine.js for lightweight reactive UI components.
- Developed secure Excel export functionalities using Maatwebsite/Excel and streamlined automated testing workflows via PHPUnit.

CRM & Biometric Attendance Module (Odoo)

- Implemented custom CRM features including automated sales pipelines, lead tracking, and seamless HRM integration with ZKTeco biometric devices.
- Built interactive dashboards and automated workflows to optimize data reporting and maximize client operational efficiency.

Website Development (Odoo)

- Designed and implemented a custom membership system within Odoo to streamline client registration, multi-tier workflows, and client updates.
- Redesigned and deployed the company website using the Odoo platform with modern navigation and mobile optimization to elevate overall user engagement.

Web-Based Management System – Fishing Pond Operations (Laravel | MySQL)

- Designed and developed a booking and inventory management platform using Laravel, HTML, CSS, and JavaScript.
- Integrated structural database management and backend logic to safely handle complex staff scheduling and real-time customer operations.

Automated Plant Watering System (Arduino | IoT | C/C++)

- Developed hardware components including soil moisture sensors, water pumps, and an Arduino microcontroller for real-time data processing and control.
- Programmed the Arduino board using C/C++ to orchestrate the operation of sensors, actuators, and communication protocols with the IoT platform.

CERTIFICATIONS

- AI For MY Future - **Microsoft**
- CompTIA Tech+ - **CompTIA**
- CCNAv7: Switching, Routing, and Wireless Essentials - **Cisco**
- CCNP Enterprise: Core Networking - **Cisco**
- Python Essentials 1 - **Cisco**
- Introduction to Cybersecurity - **Cisco**
- JavaScript Essentials 1 - **Cisco**

ACTIVITIES

- **Competitor, Curtin Malaysia CTF Hackathon (2023):** Competed in a nationwide cybersecurity capture-the-flag challenge and ranked 9th out of 88 teams across cryptography, reverse engineering, and web vulnerabilities.
- **Development Team Member, Service @ Kampung Padang Jawa (2024):** Led the engineering of an Arduino-based IoT prototype for flood prevention, coordinating hardware assembly and field testing to meet community needs.
- **AJK for SULAM Project Network Security @ SMK Seksyen 18 (2023):** Coordinated the "Remaja Bijak Siber" program, conducting interactive technical workshops on cybersecurity awareness, network safety, and digital ethics for youth.

TECHNICAL SKILLS & LANGUAGES

- **Core Programming Languages:** PHP, JavaScript, TypeScript, Python, Java, C, C++
- **Frontend Development:** Blade, Alpine.js, React, Vue.js, Inertia.js, Livewire, Tailwind CSS, Bootstrap 5, HTML5, CSS3
- **Backend Frameworks & Environments:** Laravel, Node.js, Express, NestJS, Vanilla PHP, Laravel Fortify/Breeze
- **Databases & Architecture:** PostgreSQL (v18), MySQL, Geoserver, OpenLayers (v10.7), Proj4js, Relational Data Modeling
- **DevOps, Tools & Methodology:** Git, GitHub, GitLab CI/CD, Spatie (ACL & Logs), Odoo ERP, Jira (Agile/Scrum), Unit Testing (PHPUnit)
- **Networking & Systems Security:** Network Design & Troubleshooting (TCP/IP, DNS, BGP), Cisco Device Configuration, Packet Analysis (Wireshark), Cisco Packet Tracer, GNS3, Cybersecurity Awareness, Capture The Flag (CTF) Technical Challenges

REFERENCES

1. **Madani IT Experts Sdn Bhd**
 - Noor Haliza Binti Mohamed Saufi | IT Software Executive
 - +013-328 2356
 - liza@madani.com.my
2. **Aisling Consultant (K-Youth Program)**
 - Vamanissri A/P Vasuthevan | HR
 - +010-232 0227
 - vamanissri.v@aislingpd.com
3. **Novutal Consulting Sdn Bhd**
 - Muhammad Farhaan Bin Hamdan | Manager
 - +017-321 1416
 - mfh@novutal.com